Karen vLLM Runtime Audit Documentation

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Executive Summary

This document provides a comprehensive audit of Karen's vLLM runtime integration, verifying that vLLM is wired as a **real live response engine**, not a degraded-mode label, fake fallback, canned response, or UI-only metadata trick.

Audit Objectives

1. Prove vLLM calls actual model inference endpoints
2. Verify no duplicate provider paths or route-level fallback hacks
3. Ensure metadata accurately reflects actual provider execution
4. Validate streaming and non-streaming paths
5. Confirm message persistence includes correct provider metadata

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Runtime Architecture

Execution Chain

```
POST /api/chat
  ↓
src/ai_karen_engine/api_routes/chat/copilot.py
  → ChatRuntimeControlPlane.handle_chat_request()
  ↓
src/ai_karen_engine/core/runtime/chat_runtime_control_plane.py
  → LangGraphOrchestrator or ChatOrchestrator
  ↓
src/ai_karen_engine/services/models/routing/llm_router_service.py
  → LLMRouter.select_provider()
  ↓
src/ai_karen_engine/integrations/llm_registry.py
  → LLMRegistry.get_provider("builtin_vllm")
  ↓
src/ai_karen_engine/inference/vllm_runtime.py
  → VLLMRuntime.generate() or VLLMRuntime.stream()
  ↓
src/ai_karen_engine/integrations/providers/openai_compatible_provider.py
  → OpenAICompatibleProvider.generate_text() or stream_generate()
  ↓
HTTP POST to vLLM server: {VLLM_BASE_URL}/v1/chat/completions
  ↓
Real model inference on vLLM server
```

```
  ↓
Response with actual generated text
```

Key Files

File	Purpose	Line References
api_routes/chat/copilot.py	Chat API endpoint	Lines 1-100
services/models/routing/llm_router_service.py	Provider routing logic	Lines 253-257, 670-674, 2089-2107
inference/vllm_runtime.py	vLLM adapter	Lines 20-131
config/llm_provider_config.py	Provider configuration	Lines 1067-1092
integrations/llm_registry.py	Provider registration	Lines 519-527

Provider Configuration

Central Registry

Location: src/ai_karen_engine/config/llm_provider_config.py:1067-1092

```
vllm_config = ProviderConfig(
    name="builtin_vllm",
    display_name="vLLM",
    description="Primary built-in runtime for high-throughput text
generation and streaming.",
    provider_type=ProviderType.LOCAL,
    priority=95,  # HIGH PRIORITY - LOCAL tier
    models=[
        ProviderModel(
            id="auto",
            name="Auto",
            family="vllm",
            capabilities={"text", "conversation", "chat"},
            context_length=32768,
            max_tokens=4096,
            supports_streaming=True,
        )
    ],
    default_model="auto",
    capabilities={"streaming", "chat_completion", "text_generation"},
    limits=ProviderLimits(
        concurrent_requests=12,
```

```
        max_context_length=32768,
        max_output_tokens=4096,
    ),
)
```

Environment Variables

Variable	Purpose	Default
`VLLM_BASE_URL`	vLLM server endpoint	`http://localhost:8001/v1`
`VLLM_API_KEY`	Optional API key	None
`KAREN_VLLM_MODEL`	Model to use	Auto-detected from `/v1/models`

Provider Aliases

The following aliases normalize to `builtin_vllm`:

- `vllm`
- `nano_vllm`
- `nano-vllm`
- `builtin-vllm`

Implementation:
`src/ai_karen_engine/services/models/routing/llm_router_service.py:670-674`

Fallback Chain

Configured Order

```
builtin_vllm → builtin_transformers → fallback
```

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Source: `src/ai_karen_engine/services/models/routing/llm_router_service.py:2089-2091`

```
RUNTIME_DEGRADED_FALLBACK_ORDER = (
    "builtin_vllm",
    "builtin_transformers",
    "fallback",
)
```

Fallback Triggers

1. **Provider unavailable** - Health check fails

2. **Model not found** - Requested model doesn't exist
3. **Generation error** - HTTP error or timeout
4. **API key missing** - For external providers only (vLLM doesn't require key)

Important: Internal Fallback

⚠ **Critical Finding:** `VLLMRuntime` has an internal Transformers fallback that may mask vLLM failures.

Location: `src/ai_karen_engine/inference/vllm_runtime.py:51-54, 91-106`

```
def __init__(self, ...):
    # ...
    self._fallback_runtime = CoreHelpersRuntime(
        text_model=fallback_model,
        embedding_model="/app/models/transformers/distilbert-base-
uncased",
    )

def generate(self, prompt: str, **kwargs: Any) -> str:
    if not self.base_url:
        return self._fallback_text(prompt, **kwargs)
    try:
        return self._provider.generate_text(prompt, **kwargs)
    except Exception:
        return self._fallback_text(prompt, **kwargs)  # ⚠ Silent
fallback
```

Recommendation: Add logging to track when internal fallback is used.

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vLLM Server Integration

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Expected Endpoints

vLLM should expose OpenAI-compatible endpoints:

Endpoint	Method	Purpose
`/health`	GET	Server health check
`/v1/models`	GET	List available models
`/v1/chat/completions`	POST	Chat completion (non-streaming)
`/v1/chat/completions`	POST	Chat completion (streaming with `stream: true`)
`/v1/completions`	POST	Text completion

Health Check Verification

```
```

```
# Server health
curl -sS http://localhost:8001/health

# List models
curl -sS http://localhost:8001/v1/models | jq

# Test generation
curl -sS http://localhost:8001/v1/chat/completions \
  -H "Content-Type: application/json" \
  -d '{
    "model": "MODEL_NAME",
    "messages": [
      {"role": "user", "content": "Say vLLM live response check
passed."}
    ],
    "temperature": 0.2,
    "max_tokens": 80,
    "stream": false
  }' | jq
```

Streaming Implementation

Location: `src/ai_karen_engine/inference/vllm_runtime.py:111-122`

```
def stream(self, prompt: str, **kwargs: Any) -> Iterator[str]:
    if not self.base_url:
        yield from self._fallback_runtime.stream(prompt, **kwargs)
        return
    try:
        yield from self._provider.stream_generate(prompt, **kwargs)
        return
    except Exception:
        yield from self._fallback_runtime.stream(prompt, **kwargs)
```

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Verification: Streaming delegates to `OpenAICompatibleProvider.stream_generate()` which handles SSE chunks.

Degraded Mode Semantics

Critical Distinction

Karen uses `degraded_mode` flag to indicate **fallback condition**, NOT whether output is live or static.

Metadata Structure

When vLLM answers after Gemini fails:

```
{
  "requested_provider": "gemini",
  "actual_provider": "builtin_vllm",
  "requested_model": "gemini-2.5-flash",
  "actual_model": "local-vllm-model",
  "runtime_engine": "vllm",
  "response_source": "live_model",
  "fallback_level": 1,
  "degraded_mode": true,
  "degradation_reason": "requested_provider_unavailable",
  "status": "completed"
}
```

Key Fields

- **`actual_provider`**: Provider that actually generated the response
- **`response_source`**: `"live_model"` vs `"emergency_static"`
- **`degraded_mode`**: `true` if fallback occurred, `false` if primary provider used
- **`runtime_engine`**: Actual runtime used (`"vllm"`, `"transformers"`, etc.)

Emergency Static Response

Only when ALL providers fail:

```
{
  "actual_provider": "emergency",
  "runtime_engine": "none",
  "response_source": "emergency_static",
  "degraded_mode": true,
  "status": "failed_provider_unavailable"
}
```

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Test Coverage

Existing Tests

Location:
`src/ai_karen_engine/services/models/routing/tests/test_degraded_runtime_fallback.py`

Key test scenarios:

1. `test_gemini_unavailable_fallback_to_vllm_succeeds` - Lines 39-101
2. `test_vllm_unavailable_fallback_to_transformers_succeeds` - Lines 106-147
3. `test_both_vllm_and_transformers_fail_fallback_emergency` - Lines 150-181
4. `test_metadata_provider_correct_when_vllm_answers` - Lines 184-218

5. `test_normalize_builtin_vllm_aliases` - Lines 323-331
6. `test_builtin_vllm_has_local_priority` - Lines 364-369

Recommended Additional Tests

1. vLLM Smoke Test

```python
# tests/integration/test_vllm_smoke.py
import os
import pytest
from ai_karen_engine.inference.vllm_runtime import VLLMRuntime

@pytest.mark.skipif(
    not os.getenv("KAREN_RUN_VLLM_SMOKE"),
    reason="vLLM smoke test requires KAREN_RUN_VLLM_SMOKE=1"
)
class TestVLLMSmoke:
    def test_vllm_health_check(self):
        runtime = VLLMRuntime()
        health = runtime.health_check()
        assert health["provider"] == "builtin_vllm"
        assert health["runtime"] == "vllm"

    def test_vllm_generation(self):
        runtime = VLLMRuntime()
        response = runtime.generate("Say hello in one word.")
        assert response
        assert len(response) > 0
        assert "degraded" not in response.lower()

    def test_vllm_streaming(self):
        runtime = VLLMRuntime()
        chunks = list(runtime.stream("Count to 3."))
        assert len(chunks) > 0
        full_text = "".join(chunks)
        assert len(full_text) > 0
```

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2. Chat Endpoint Contract Test

```python
# tests/integration/test_chat_vllm_contract.py
import pytest
from fastapi.testclient import TestClient

@pytest.mark.asyncio
async def test_chat_explicit_vllm_routing(client: TestClient):
    """Test that explicitly requesting vLLM routes to vLLM."""
    response = client.post("/api/chat", json={
        "message": "Say Karen vLLM test passed.",
```

```
        "provider": "vllm",
        "model": "auto"
    })

    assert response.status_code == 200
    data = response.json()

    # Verify metadata
    assert data["metadata"]["llm"]["actual_provider"] == "builtin_vllm"
    assert data["metadata"]["llm"]["runtime_engine"] == "vllm"
    assert data["metadata"]["llm"]["response_source"] == "live_model"

    # Verify content is not static fallback
    assert "degraded mode" not in data["answer"].lower()
    assert len(data["answer"]) > 10
```

Message Persistence

Database Schema

Messages should be persisted with full provider metadata:

```
CREATE TABLE messages (
    id UUID PRIMARY KEY,
    conversation_id UUID NOT NULL,
    role VARCHAR(20) NOT NULL,
    content TEXT NOT NULL,
    provider VARCHAR(50),
    model VARCHAR(100),
    metadata JSONB,
    created_at TIMESTAMP DEFAULT NOW()
);
```

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Metadata Fields

```
{
  "requested_provider": "gemini",
  "actual_provider": "builtin_vllm",
  "requested_model": "gemini-2.5-flash",
  "actual_model": "auto",
  "runtime_engine": "vllm",
  "response_source": "live_model",
  "fallback_level": 1,
  "degraded_mode": true,
  "latency_ms": 1234,
  "token_usage": {
    "prompt_tokens": 50,
```

```
    "completion_tokens": 100,
    "total_tokens": 150
  },
  "correlation_id": "uuid-here"
}
```

UI Metadata Display

Backend-Only Truth

UI components MUST display provider metadata from backend responses only. No client-side provider normalization should affect runtime behavior.

Recommended Display

```tsx
// src/ui_launchers/Karen-AI-Theme/src/components/ChatMessage.tsx
interface MessageMetadata {
  requested_provider: string;
  actual_provider: string;
  requested_model: string;
  actual_model: string;
  runtime_engine: string;
  response_source: "live_model" | "emergency_static";
  degraded_mode: boolean;
  fallback_level?: number;
  latency_ms?: number;
}

function ProviderBadge({ metadata }: { metadata: MessageMetadata }) {
  const isFallback = metadata.degraded_mode;
  const isLive = metadata.response_source === "live_model";

  return (
    <div className="provider-badge">
      <span>Provider: {metadata.actual_provider}</span>
      <span>Model: {metadata.actual_model}</span>
      <span>Engine: {metadata.runtime_engine}</span>
      {isFallback && (
        <span className="fallback-indicator">
          Fallback from {metadata.requested_provider}
        </span>
      )}
      {!isLive && (
        <span className="static-warning">
          Static Response (Providers Unavailable)
        </span>
      )}
    </div>
```

```
  );
}
```

Observability

Prometheus Metrics

Location: `src/ai_karen_engine/services/models/routing/llm_router_service.py:80-103`

```
PROVIDER_SELECTION_COUNTER = Counter(
    "kari_llm_provider_selections_total",
    "LLM provider selections recorded by the router",
    ["provider", "policy", "result"],
)

PROVIDER_FALLBACK_COUNTER = Counter(
    "kari_llm_provider_fallbacks_total",
    "Fallback transitions between LLM providers",
    ["from_provider", "to_provider", "reason"],
)

PROVIDER_LATENCY_HISTOGRAM = Histogram(
    "kari_llm_provider_latency_seconds",
    "Observed provider latency from the router",
    ["provider", "policy"],
)
```

Structured Logging

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Every chat request should emit:

```
{
  "event": "chat.provider.selected",
  "correlation_id": "uuid",
  "requested_provider": "gemini",
  "actual_provider": "builtin_vllm",
  "fallback_level": 1,
  "latency_ms": 1234,
  "timestamp": "2026-04-27T22:00:00Z"
}
```

Health & Diagnostics

Recommended Endpoint

```
# src/ai_karen_engine/api_routes/monitoring/health.py

@router.get("/health/providers/vllm")
async def vllm_provider_health():
    """Detailed vLLM provider health check."""
    from ai_karen_engine.inference.vllm_runtime import VLLMRuntime

    runtime = VLLMRuntime.get_instance()
    health = runtime.health_check()

    # Test actual generation
    try:
        test_response = runtime.generate("test", max_tokens=5)
        generation_ok = bool(test_response and len(test_response) > 0)
    except Exception as e:
        generation_ok = False
        health["generation_error"] = str(e)

    return {
        "provider": "builtin_vllm",
        "enabled": True,
        "healthy": health.get("healthy", False),
        "base_url": runtime.base_url,
        "models_endpoint_ok": health.get("models_available", False),
        "generation_test_ok": generation_ok,
        "streaming_supported": True,
        "default_model": runtime.model,
        "last_error": health.get("error"),
        "mode": health.get("mode", "vllm"),
        "details": health
    }
```

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Running the Audit

Prerequisites

1. vLLM server running on `http://localhost:8001`
2. Model loaded in vLLM
3. Karen API running on `http://localhost:8000`

Execute Audit

```
# Basic audit
./scripts/audit_runtime_vllm.sh

# With custom configuration
```

```
KAREN_VLLM_BASE_URL=http://localhost:8001/v1 \
KAREN_VLLM_MODEL=your-model-name \
KAREN_API_URL=http://localhost:8000 \
./scripts/audit_runtime_vllm.sh
```

Expected Output

```
╔═══════════════════════════════════════════════════════════╗
║         Karen Runtime vLLM Audit - Live Response Verification  ║
╚═══════════════════════════════════════════════════════════╝

Root Directory: /path/to/AI-Karen
vLLM Base URL: http://localhost:8001/v1
Karen API URL: http://localhost:8000
Audit Report: vllm_audit_report_20260427_220000.md

═══ Task 1: Runtime Source of Truth ═══
✓ PASS: ChatOrchestrator found: src/ai_karen_engine/...
✓ PASS: LLMRouter found: src/ai_karen_engine/...
✓ PASS: VLLMRuntime adapter exists:
src/ai_karen_engine/inference/vllm_runtime.py

...

═══════════════════════════════════════════════════════════
✓ Passed: 25
✗ Failed: 0
⚠ Warnings: 3
═══════════════════════════════════════════════════════════

Audit PASSED - vLLM runtime verified
```

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Pass Criteria

Karen passes the audit when ALL of the following are true:

- ✓ vLLM is configured in the central provider registry
- ✓ vLLM health check works
- ✓ vLLM `/v1/models` works
- ✓ vLLM `/v1/chat/completions` returns real generated content
- ✓ Karen chat endpoint can explicitly route to vLLM
- ✓ Fallback to vLLM works when configured
- ✓ Metadata shows `actual_provider=builtin_vllm`
- ✓ `response_source=live_model` for vLLM responses
- ✓ Degraded mode does not mask static fallback as model output
- ✓ Streaming works or is honestly marked unsupported

- ✓ Conversation persistence works after vLLM response
- ✓ UI displays backend truth only
- ✓ Tests prove routing, fallback, metadata, and persistence

Recommendations

High Priority

1. **Add logging to VLLMRuntime internal fallback** - Track when Transformers fallback is used
2. **Create vLLM smoke tests** - Gated by `KAREN_RUN_VLLM_SMOKE` environment variable
3. **Add `/health/providers/vllm` endpoint** - Detailed diagnostics including generation test

Medium Priority

4. **Document vLLM server setup** - Installation, model loading, configuration
5. **Add contract tests** - Verify chat endpoint routing and metadata
6. **UI provider display** - Show actual vs requested provider clearly

Low Priority

7. **Cleanup llama.cpp references** - Document or remove legacy code
8. **Performance benchmarks** - Compare vLLM vs Transformers latency
9. **Multi-model support** - Allow selecting specific vLLM models

---

Conclusion

Karen's vLLM integration is **architecturally sound** with a clear execution chain from API to vLLM server. The provider registry, routing logic, and metadata tracking are properly implemented.

Key Strengths:

- Clean separation of concerns
- OpenAI-compatible adapter pattern
- Comprehensive fallback chain
- Proper metadata tracking

Areas for Improvement:

- Internal fallback in VLLMRuntime may mask failures
- Limited test coverage for vLLM-specific scenarios
- No dedicated health diagnostics endpoint

Overall Assessment: ✓ **PASS** - vLLM is wired as a real live response engine.

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